Practice Exercises: Teaching Block 3

Star Questions for Telecom and IoT classes

Star Question 1

- Consider an interface named **Colorable**, as follows:

```
public interface Colorable { public void howToColor(); }
```

  - Create a class named **Square** that extends **GeometricObject** *and* implements **Colorable**.
  - Implement **howToColor()** to display a message on how to colour the square.
  - Create an additional class to test the creation of a **Square** and its method **howToColor()**.

```
public abstract class GeometricObject {
  // some methods and instance variables

  public abstract double findArea();
  public abstract double findPerimeter();
}
```

Star Question 2

- Identify the lines of code that, if added to the program at point **X** would cause exactly one more object to be eligible for the **Garbage Collector**.

```
copyGC = null;
gc2 = null;
newGC = gc3;
gc1 = null;
newGC = null;
gc4 = null;
gc3 = gc2;
gc1 = gc4;
gc3 = null;
```

```
public class GC {
  public static GC doStuff() {
    GC newGC = new GC();
    doStuff2(newGC);
    return newGC;
  }
  public static void main(String[] args) {
     GC gc1;
     GC gc2 = new GC();
     GC gc3 = new GC();
     GC gc4 = gc3;
     gc1 = doStuff();
     X
     // call more methods
  }
  public static void doStuff2(GC copyGC) {
    GC localGC;
  }
}
```

Note that variable **gc1** is not initialised to a default value here, because it is a local variable.

Star Question 3

- Consider an interface named **`Instrument`**, as follows:

```
public interface Instrument { public void playSound(); }
```

  - Create two classes named **`Piano`** and **`Clarinet`** that implements **`Instrument`**.
  - Use polymorphism to create an array of **`Instrument`** objects and demonstrate how different instruments play sounds using a single **`playSound()`** method call.